

MobiLens AI — Human-Centric Mobility Intelligence & Intervention Simulator

A Complete Guide to Concept, Mathematical Engine, Architecture, and All 13 Pages

Core Paradigm: "Don't just tell people where the bus is. Tell them what that means for their entire human journey."

Conventional transit tools are vehicle-centric (tracking dots on a map). MobiLens AI is human-centric: evaluating cumulative exertion, thermal heat exposure, transfer traps, accessibility barriers, and dynamic disruption recovery.

1. Core Concept & The Human-Centric Paradigm

MobiLens AI addresses the critical missing link in modern urban transportation: the **Mobility Friction** experienced by everyday citizens, seniors, students, and persons with disabilities. While standard transit APIs track scheduled vehicle timings, commuters experience the friction of unshaded walkways, dangerous multi-lane highway transfers without crosswalks, delayed intermodal connections, and broken wheelchair access.

MobiLens AI operates on a continuous feedback loop:

1. HUMAN JOURNEY User's complete trip door-to-door (Walk → Bus → Transfer → Final Mile)	2. FRICTION AUDIT Real-time penalty scoring (Thermal heat, wait time, physical effort)	3. BOTTLENECK Detection of transfer traps & connection failure points	4. INTERVENTION AI-generated micro-fixes & policy simulation before spending capital
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2. Mathematical Formulation: Mobility Friction Index (MFI)

The **Prototype Mobility Friction Index (MFI)** is a calibrated scalar index ranging from **0 (Frictionless)** to **100 (Critical Strain)**:

MFI Equation:

$$\text{MFI} = w_{\text{TT}} \cdot (\text{Travel Time}) + w_{\text{WB}} \cdot (\text{Waiting Burden}) + w_{\text{TR}} \cdot (\text{Transfer Risk}) + w_{\text{HM}} \cdot (\text{Heat Multiplier}) + w_{\text{AB}} \cdot (\text{Accessibility Barriers})$$

Factor	Weight / Penalty	Behavioral & Empirical Justification
Wait Burden	2.5× base time	Commuter perception studies demonstrate that waiting at unsheltered, unseated bus stops feels 2.5× longer than in-vehicle travel time.
Transfer Trap Risk	+7 to +20 MFI	Penalizes multi-lane highway road crossings with high vehicular speeds, missing crosswalk signals, or unpaved shoulders.
Heat Stress Multiplier	1.25× to 1.6×	Calculated dynamically from real-time ambient temperature (>32°C), direct midday sun exposure, and lack of tree canopy/shade.
Accessibility Barrier	+35 to +60 MFI	Applied when steep ramp gradients (>1:12), broken curbs, or non-wheelchair-accessible high-floor buses are present on a route.

3. Technical Stack & Deployment Architecture

Layer	Technologies	Key Responsibilities & Integrations
Frontend UI	React 18, TypeScript, Vite, Tailwind CSS, Lucide Icons	Zero-lag reactive user interface, glassmorphism design system, responsive mobile & desktop viewports.
GIS Mapping	Leaflet, React-Leaflet, OpenStreetMap Standard Tiles	High-performance geospatial visualization, interactive polyline corridors, custom DivIcon bus capsules, zero watermark tiles.
Backend API	Python, FastAPI, Uvicorn, WebSockets	RESTful endpoints, real-time vehicle coordinate interpolation, GTFS/GTFS-RT abstraction layer, risk engines.
Database	MongoDB Atlas Cloud Database (Cluster0)	Persistent storage for user profiles, crowdsourced transfer traps, saved custom journeys, and historical city audits.

4. Complete Page-by-Page Deep Dive (All 13 Modules)

The MobiLens AI suite features 13 purpose-built modules designed for citizens, planners, administrators, and accessibility advocates:

Page 1: Overview Dashboard (`/overview`) — Executive City Mobility Health & District Comparisons

Provides an aggregated city-wide view of Mobility Friction across Tamil Nadu districts (Tirunelveli, Chennai, Coimbatore, Madurai, Salem, etc.). Features a real-time MFI speedometer, district comparative cards, and a live ticker tracking active bus dispatches, citizen transfer trap reports, and high-heat corridor alerts.

Page 2: My Journey (`/my-journey`) — Personalized Multimodal Commute Intelligence

Allows individual citizens to plan and inspect their door-to-door commute broken down into walk, bus, transfer, and feeder legs. Features the visual Heat Stress & Sun Exposure multiplier (fatigue meter) and the crowdsourced 'Report Transfer Trap' modal that directly feeds into the city planner database.

Page 3: Live Journey & Bus Tracking (`/live-journey`) — Contextual Real-Time Corridor Tracking & Disruption Recovery

The flagship feature of MobiLens AI. Focuses specifically on the user's active travel corridor (Thoothukudi Airport TCR → Francis Xavier Engineering College Vannarpettai on NH 138) with Bus 15 and Feeder 7B-EV. Features the transparent 'Can I Catch This Bus?' modal, 1-click delay simulation (+6 min), and 4 AI recovery alternatives.

Page 4: Mobility Friction Map (`/mobility-map`) — Spatial GIS Heatmaps & Friction Corridors

Interactive geographic mapping layer visualizing transfer traps, high-friction bus stops, pedestrian choke points, and micro-bottlenecks. Planners can toggle layers for heat stress, accessibility, and stop crowding.

Page 5: Journey Analyzer (`/journey-analyzer`) — Deep Forensics on Route Exertion & Delay

Performs granular forensics on every meter of a planned route. Deconstructs the friction score into physical walking exertion, shelter deficits, crossing risks, and schedule desynchronization.

Page 6: AI Interventions (`/interventions`) — Algorithmic Micro-Interventions & ROI Analysis

Recommends targeted municipal interventions to eliminate bottlenecks: green pedestrian shaded corridors, high-frequency low-floor EV shuttles, smart crosswalk signals, and sheltered bus bays, complete with estimated MFI reduction and CAPEX ROI.

Page 7: What-If Simulator (`/simulator`) — Interactive Urban Planning Policy Sandbox

Enables transport planners to drag policy sliders (e.g. increase feeder frequency by 50%, add sidewalk shading, introduce dedicated bus lanes) and immediately simulate the city-wide impact on citizen time saved, carbon reduction, and friction drops before allocating capital.

Page 8: Accessibility Navigator (`/accessibility`) — Inclusive Step-Free & Wheelchair Route Intelligence

Dedicated navigation suite prioritizing step-free paths, tactile paving, curb-cut ramp availability (<1:12 slope), auditory signals, and wheelchair-accessible low-floor bus fleets (Route 7B-EV).

Page 9: Citizen Safe Routes (`/safe-routes`) — Pedestrian & Women's Night Safety Routing

Evaluates pedestrian corridors based on human safety parameters: street lighting continuity, crowd density, open commercial storefront activity, and proximity to emergency kiosks and police outposts.

Page 10: City Intelligence Command Center (`/city-intelligence`) — Municipal Transport Operations & Bottleneck Spotlight

Enterprise command center for city transport corporations. Features the Live Transit Corridor Bottleneck Spotlight, tracking route delays, citizen trap submissions, and converting bottlenecks into municipal work orders.

Page 11: Mobility Analytics (`/analytics`) — Long-Term Trends & Modal Shift Reports

Aggregates macro-level urban mobility metrics: average commuter delay distributions, seasonal heat impact trends, modal shift from private vehicles to public transit, and municipal carbon savings.

Page 12: Pitch Deck & Vision (`/about`) — Executive Narrative & Problem-Solution Canvas

Comprehensive presentation canvas detailing the economic burden of transit friction, human stories of commuting barriers, technical architecture, and scalability across global metropolitan regions.

Page 13: Persona Authentication (`/login`) — Role-Based Access for 4 Distinct Stakeholders

Configures the entire workspace dynamically based on 4 persona roles: Daily Commuter / Citizen, Urban Transport Planner, Accessibility Advocate / Wheelchair User, and City Administrator.

5. Flagship Capability: The Live Journey Showcase

The Live Journey module represents the synthesis of live transit telemetry and human-centric journey intelligence. Instead of an overwhelming generic map with hundreds of congested buses, MobiLens AI displays an ultra-clean corridor tailored to the user's active commute:

Active Showcase Corridor: NH 138 Inter-District Highway

- **Origin:** Thoothukudi Airport (TCR) Terminal (Lat: 8.7242, Lng: 78.0264)
- **Intermediate Hubs:** Vagaikulam Feeder Stop, Vallanadu Junction, Thamirabarani River Bridge
- **Destination:** Francis Xavier Engineering College (Fxec) Main Gate, Vannarpettai, Tirunelveli
- **Assigned Vehicles:** Bus 15 (Highway Express) & Feeder 7B-EV (Zero-Emission Campus Connector)

A. "Can I Catch This Bus?" Transparent Arithmetic Calculator

Instead of a mysterious "Yes/No" indicator, MobiLens AI displays the exact transparent calculation so citizens feel empowered and in control:

Parameter	Value & Unit	Description in Transparent Arithmetic
Walking Distance	280 meters	Direct Haversine pedestrian distance from user's origin to bus bay
Walking Speed	4.5 km/h	Standard pedestrian walking speed (adjusted to 3.2 km/h for wheelchair mode)
Weather Multiplier	1.25× (Midday Sun)	Thermal heat factor slowing physical walking velocity
Required Walking Time	4.2 minutes	Gross walking duration including safe road crossing buffer
Bus Arrival ETA	6.0 minutes	Live interpolated GPS countdown of approaching Bus 15
Safety Buffer Margin	+1.8 minutes	Positive buffer confirms user will comfortably catch the vehicle

B. Disruption & 4 Dynamic AI Recovery Alternatives

When unexpected traffic delays Bus 15 by +6 minutes, the system immediately recognizes that the transfer at the highway junction will fail. The algorithm dynamically recalculates and presents 4 recovery alternatives:

Alternative Option	Strategy & Transfer Mechanism	ETA / Friction / Risk
Option A Stay on Delayed Bus	Accept delay and wait for subsequent connecting feeder service at junction.	ETA: +18m late Friction: 68 (High) Risk: High
Option B Rapid Bypass Shuttle	Board passing express shuttle at highway junction to bypass local city stop delays.	ETA: On Time (08:52) Friction: 38 (Low) Risk: Low
Option C (Recommended) Campus Direct EV Feeder	Hop onto direct electric feeder linking airport terminal directly to Fxec campus gate.	ETA: 12m early Friction: 24 (Smooth) Risk: Low
Option D Reroute to Vallanadu Hub	Transfer at the sheltered Vallanadu depot rather than the highway shoulder.	ETA: +6m late Friction: 46 (Moderate) Risk: Moderate

6. User Personas & Tailored Workflow Matrix

Persona Role	Target User Group	Default Features & Value Delivered
Daily Commuter	College students, office workers, everyday citizens	My Journey planning, live bus countdowns, weather fatigue alerts, 1-click Transfer Trap reporting.
Urban Planner	Transit authorities, municipal engineers, traffic consultants	What-If Simulation sandbox, policy testing, corridor bottleneck spotlights, AI intervention ROI models.
Accessibility User	Wheelchair users, senior citizens, visually impaired	Step-free routing, curb-cut ramp audits, wheelchair-accessible low-floor bus dispatch filters (Route 7B-EV).
City Administrator	Transport commissioners, city mayors, transit directors	City Intelligence Command Center, aggregate district MFI gauges, work-order dispatch for citizen trap reports.

7. Quality Assurance & System Verification

- **Production TypeScript Build:** Fully verified with `npx tsc --noEmit` (0 compilation errors).
- **Map Cleanliness:** OpenStreetMap standard tiles deployed with zero API-key watermarks and de-cluttered corridor bus filtering.
- **Real-Time Backend Services:** FastAPI Uvicorn engine running at `http://localhost:8000` with live MongoDB Atlas connectivity.
- **Interactive Frontend:** Vite reactive development application serving at `http://localhost:5173`.

MobiLens AI — Empowering human mobility through actionable, human-centered intelligence.
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